Real Numbers (Set A) — MCQ Worksheet

Mathematics • Chapter 1 • *25 Questions, 1 mark each — Total: 25 marks*

Instructions: *Choose the correct option (A / B / C / D) for each question. Each question carries 1 mark. There is no negative marking. Suggested time: 30 minutes.*

Q1. The HCF of two numbers a and b is the:

(A) Largest number that divides both
(B) Smallest number
(C) Difference of a and b
(D) Sum of a and b

Q2. Euclid's division lemma states that for any two positive integers a, b there exist unique integers q, r such that $a = bq + r$ where:

(A) $0 \le r < b$
(B) $r > b$
(C) $r < 0$
(D) $r = b$

Q3. HCF (96, 404) is:

(A) 2
(B) 4
(C) 8
(D) 16

Q4. LCM × HCF of two numbers a, b equals:

(A) $a + b$
(B) $a – b$
(C) $a \times b$
(D) a/b

Q5. By Fundamental Theorem of Arithmetic, every composite number can be factorised as a product of:

(A) Two primes only
(B) Primes (uniquely up to order)
(C) Two even numbers
(D) Three primes

Q6. The prime factorisation of 156 is:

(A) $2^2 \times 3 \times 13$
(B) $2 \times 3^2 \times 13$
(C) $2^2 \times 39$
(D) 2×78

Q7. LCM of 15 and 20 is:

(A) 20
(B) 60
(C) 100
(D) 300

Q8. HCF of 24 and 36 is:

(A) 6
(B) 12
(C) 18
(D) 24

Q9. $\sqrt{2}$ is:

(A) A rational number
(B) An irrational number
(C) An integer
(D) A natural number

Q10. $5 + \sqrt{3}$ is:

(A) Rational
(B) Irrational
(C) Integer
(D) Natural

Q11. A rational number p/q ($q \neq 0$) has a terminating decimal expansion if q (in lowest form) has prime factors:

(A) Of the form 2■ × 5■ only
(B) Includes 3
(C) Includes 7
(D) Includes 11

Q12. The decimal expansion of 13/3125 is:

(A) Non-terminating recurring
(B) Terminating
(C) Non-terminating non-recurring
(D) Irrational

Q13. The decimal expansion of 23/8 is:

(A) Terminating
(B) Non-terminating recurring
(C) Non-terminating non-recurring
(D) Irrational

Q14. The decimal expansion of 17/30 is:

(A) Terminating
(B) Non-terminating recurring
(C) Irrational
(D) Non-real

Q15. The product of two irrational numbers is:

(A) Always rational | (B) Always irrational
(C) Sometimes rational, sometimes irrational | (D) Always zero

Q16. $3 \times \sqrt{7}$ is:

(A) Rational | (B) Irrational
(C) Integer | (D) Natural

Q17. If HCF (a, b) = 5 and LCM (a, b) = 200, then $a \times b$ is:

(A) 100 | (B) 200
(C) 1000 | (D) 500

Q18. LCM (8, 9, 25) is:

(A) 1800 | (B) 1500
(C) 800 | (D) 2250

Q19. The smallest composite number is:

(A) 1 | (B) 2
(C) 4 | (D) 6

Q20. The smallest prime number is:

(A) 1 | (B) 2
(C) 3 | (D) 5

Q21. The HCF of two consecutive integers is:

(A) 0 | (B) 1
(C) 2 | (D) Their sum

Q22. The HCF of two consecutive even integers is:

(A) 1 | (B) 2
(C) 4 | (D) Their product

Q23. If p is a prime and p divides a^2, then p divides:

(A) a | (B) a^2
(C) 2a | (D) Always 0

Q24. Numbers like 0.10110011100011110… are:

(A) Rational | (B) Irrational
(C) Integers | (D) Natural

Q25. The exponent of 2 in the prime factorisation of 144 is:

(A) 2 | (B) 3
(C) 4 | (D) 5

— End of Worksheet —